Electrical Structural Health Monitoring of Cu-Mesh-Integrated CFRP Adhesive Joints

[Author names and order to be confirmed] • [Corresponding author]

[Department / Institution] | [Contact email]

[Conference year / session / poster number]

Introduction

Can an embedded copper mesh and the CFRP conductive network provide an electrical indicator of adhesive-joint damage?

Combine directional resistance measurements with single-lap shear (SLS) testing and post-test surface observations.

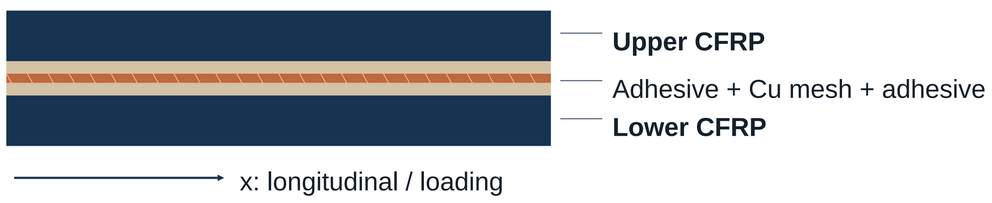
BASELINE — Map current paths → **JOINT** — Integrate Cu mesh → **SHM** — Track R during SLS

Experimental Setup & Sensing Concept

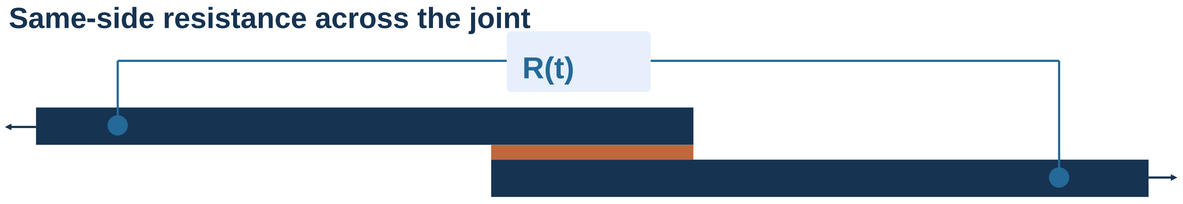
CFRP lay-up: $[0/45/90/-45]_{2s}$

40-mesh Cu: principal direction at 90° to loading x.

Joint cross-section (schematic)



Same-side resistance across the joint



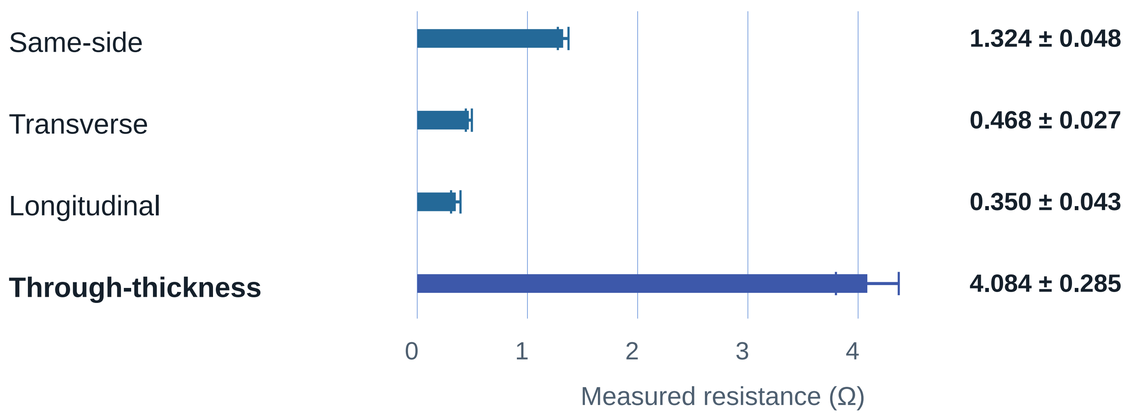
Diagrams are conceptual and not to scale.

Working hypothesis: competing current paths

CFRP paths, Cu-mesh bypasses and local contacts form a network. Interface separation or contact loss can change R; the mesh is not a single series resistor.

Electrical Characterization

Single laminate: mean ± sample SD (Ω)



Cu-mesh joint: measured resistance (Ω)

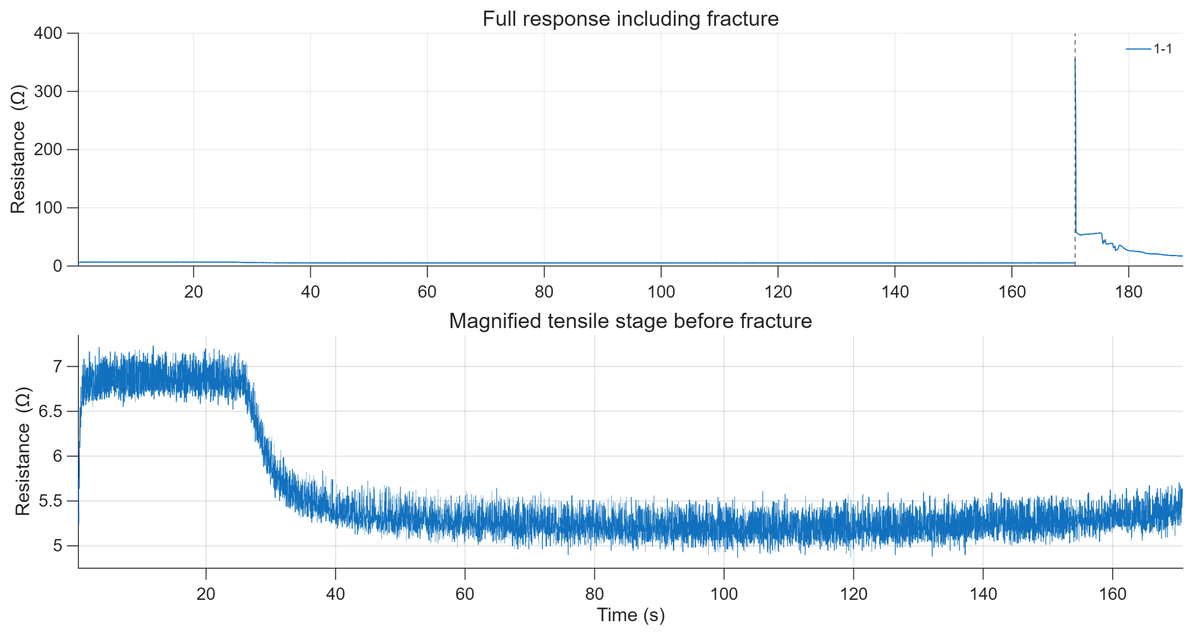
Configuration	Mean ± sample SD
Longitudinal	**3.295 ± 0.525**
Through-thickness	**4.331 ± 0.695**
Same-side	**5.711 ± 0.841**

Each row uses 5 listed measurements; independence / repeat structure remains to be confirmed. Statistics recalculated from source tables [1].

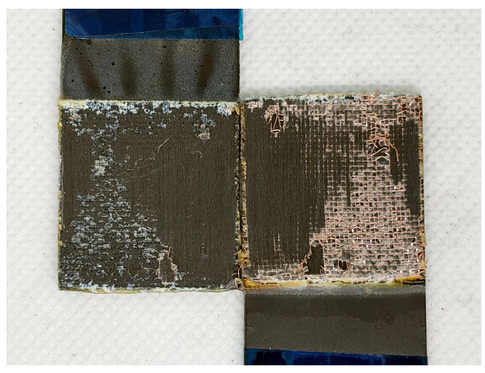
Interpretation: electrode configuration and current-path geometry must be controlled when comparing R.

Results: SHM Response

Representative specimen 1-1: supplied R(t) trace [1]



The large event-related resistance peak contrasts with the smaller, non-monotonic pre-fracture evolution. A declining R before failure is also observed.



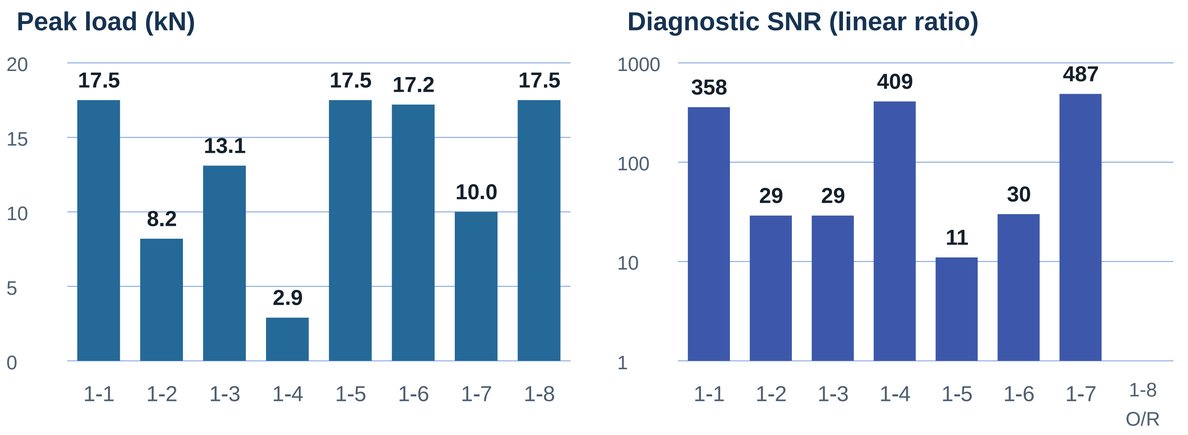
Post-test surfaces: 1-1

Visible exposed mesh and separated joint surfaces are consistent with conductive-path reconfiguration. Failure mode requires systematic classification.

Original trace and photograph retained; no time-series data reconstructed.

Results: Specimen Variability

Peak load and reported diagnostic SNR [1]



SNR = |ΔRdurable| / σbaseline • log axis • O/R: over-range

13.0 ± 5.5 kN	≈42% CV	11–487
Peak load; sample SD	Mechanical scatter	Reported finite SNR

Charts use rounded values printed in the source summary. SNR was not recomputed; over-range is excluded from its numeric range. Peak load is not lap-shear strength.

Conclusions & Future Work

- Electrical monitoring shows potential as a fracture-event indicator in Cu-mesh CFRP joints.
- Signal magnitude varies strongly; high diagnostic SNR does not establish high load capacity.
- Next: synchronize raw signals, classify failure modes and validate noise thresholds and lead time.

A fracture-related electrical event is observed; pre-fracture resistance trends vary across specimens.

8 SLS specimens

11.7× Measured R ratio (through / longitudinal)

[1] SHM-v4.pptx: pp. 2, 11–12, 22–29, 33, 38–39. Primary project material.

[Funding / acknowledgements / selected literature to be added]